MUSHIN

System Diagrams

Companion technical documentation for the MUSHIN SRS

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13 diagrams covering system context, architecture, data, workflows, sequences, and deployment

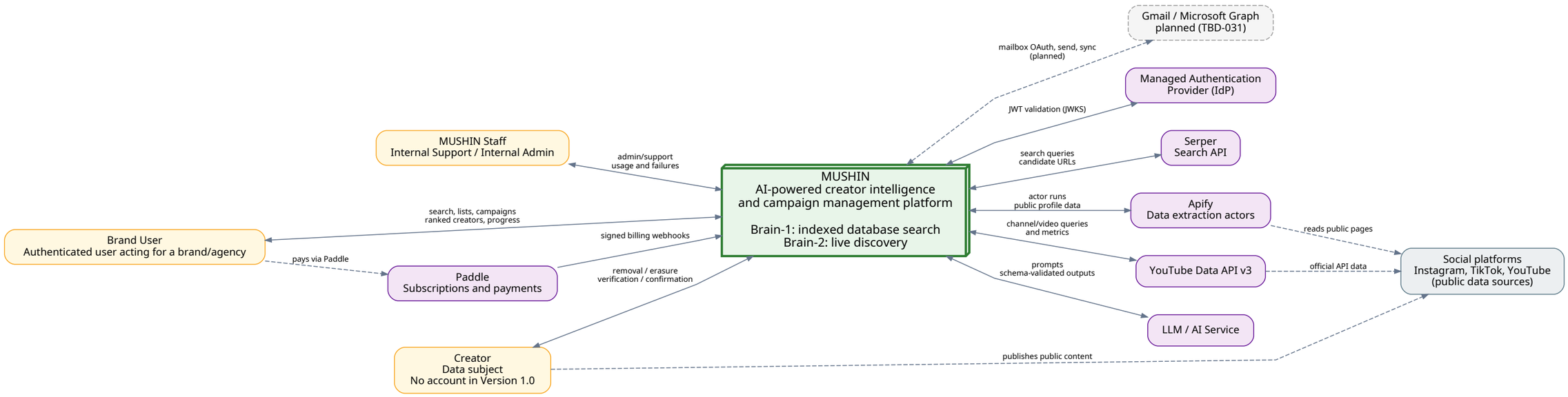
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Diagram Consistency Notes — NOTES

1. System Context Diagram

MUSHIN and its primary actors and external service integrations.

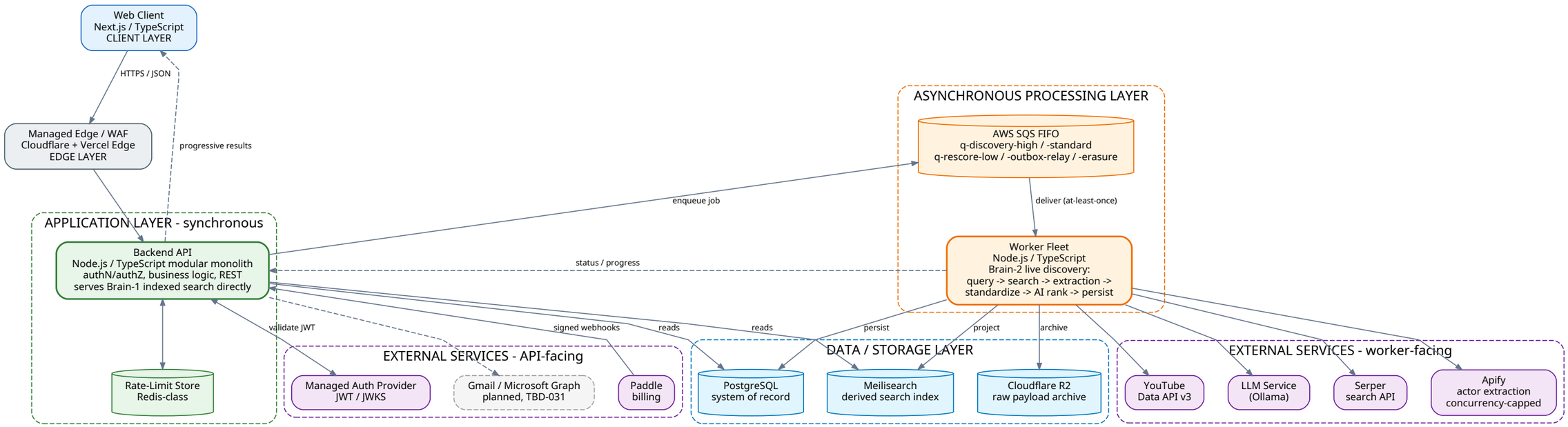


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Diagram rendered as vector artwork for readability

2. System Architecture Diagram

Layered technical architecture: client, edge, application, asynchronous processing, storage, and external services.

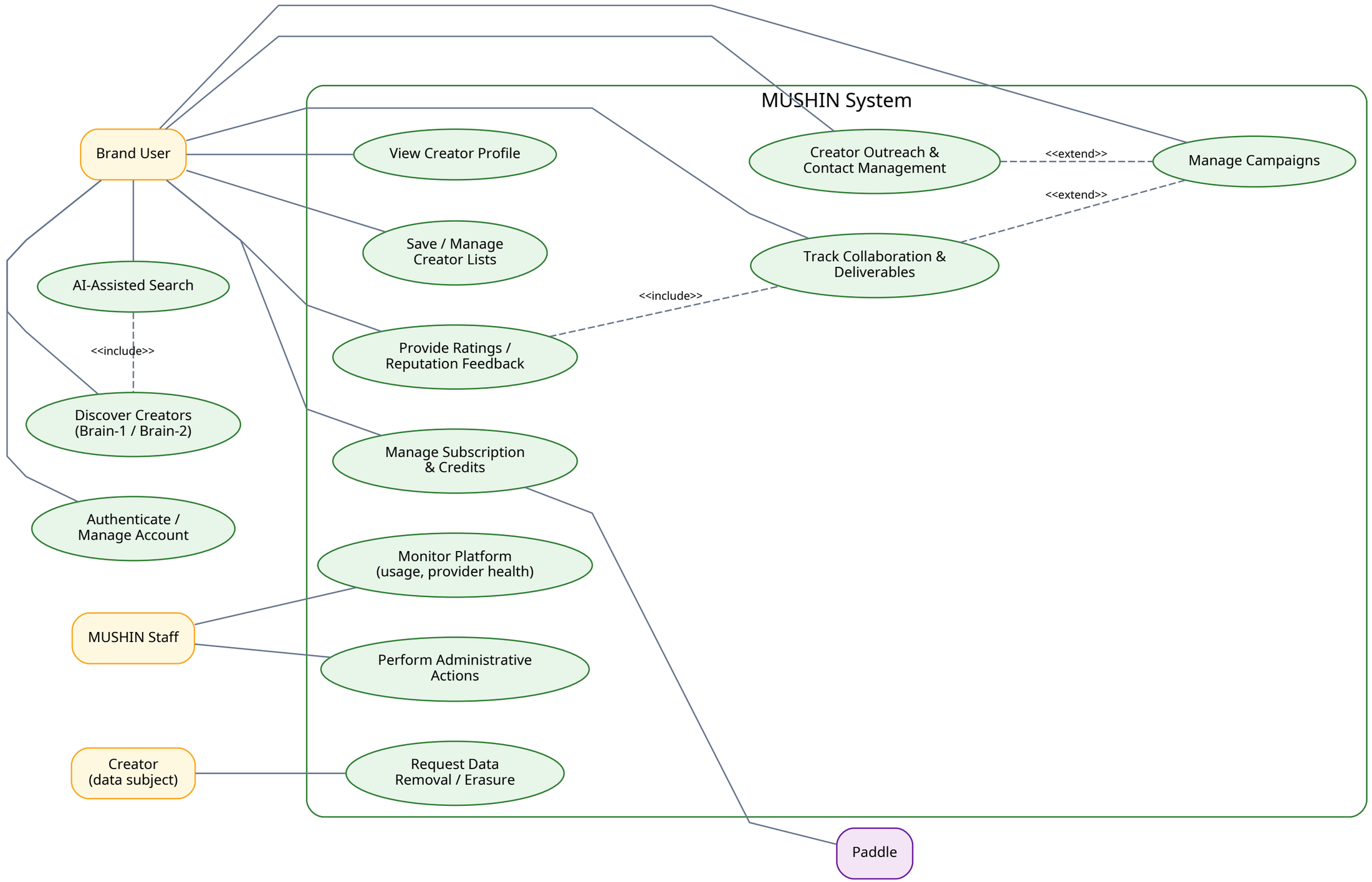


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3. Use Case Diagram

Major user and external-actor interactions with the MUSHIN system.

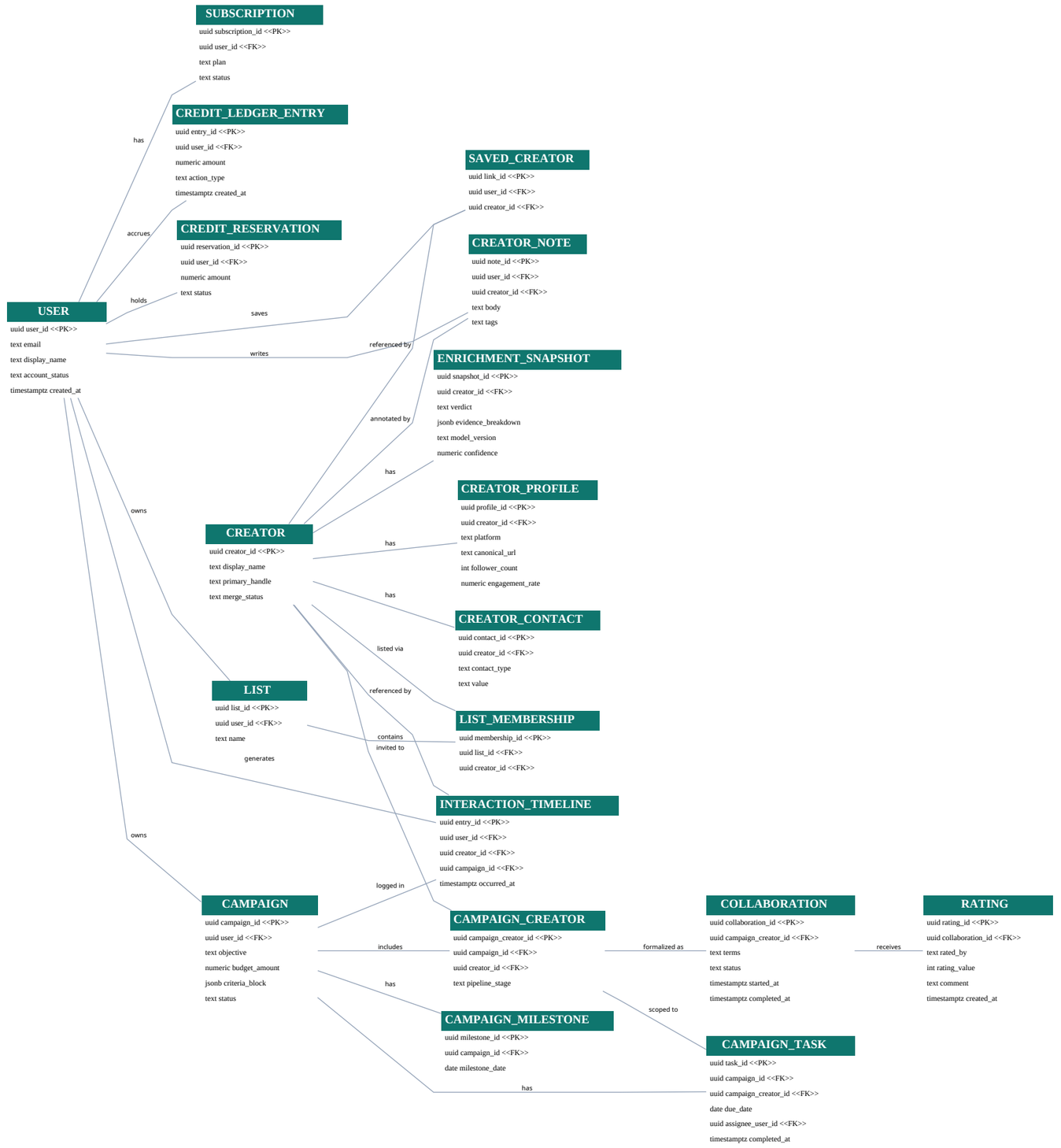


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4. Entity Relationship Diagram

Global creator data and user-owned operational data, with primary/foreign keys shown.

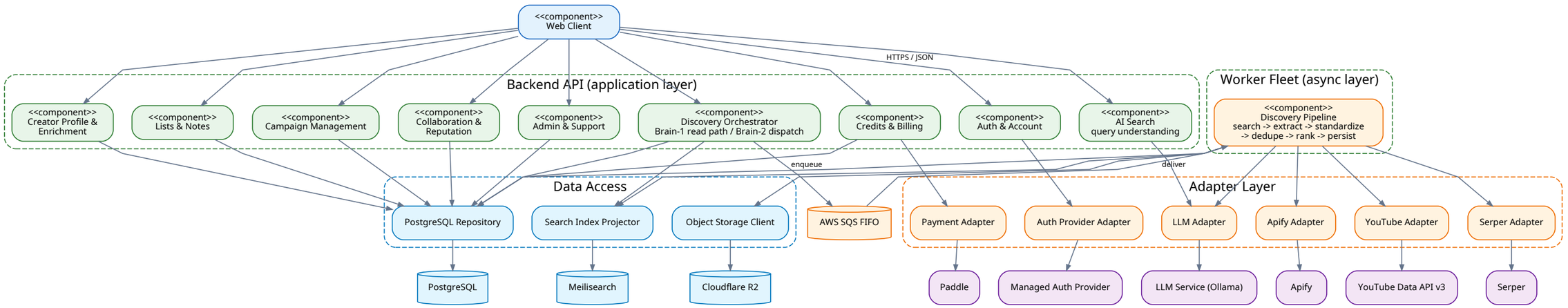


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5. Component Diagram

Logical components, adapters, data access, and external dependencies.

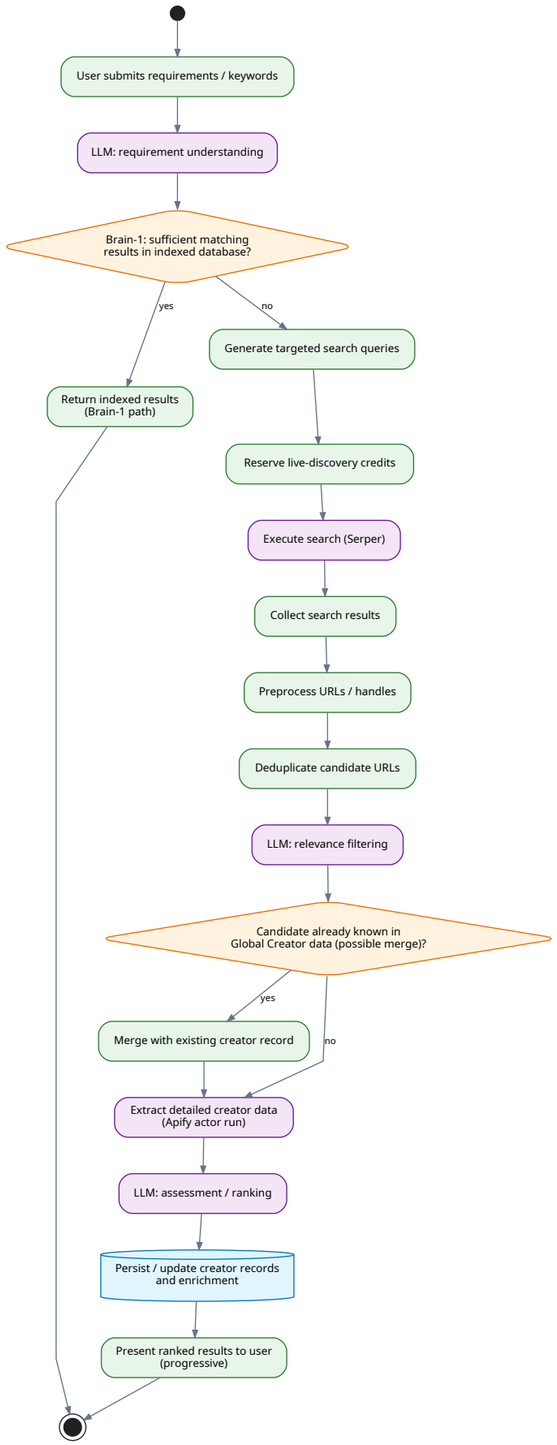


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6. Creator Discovery Activity Diagram

Brain-1 indexed retrieval and Brain-2 live-discovery workflow.

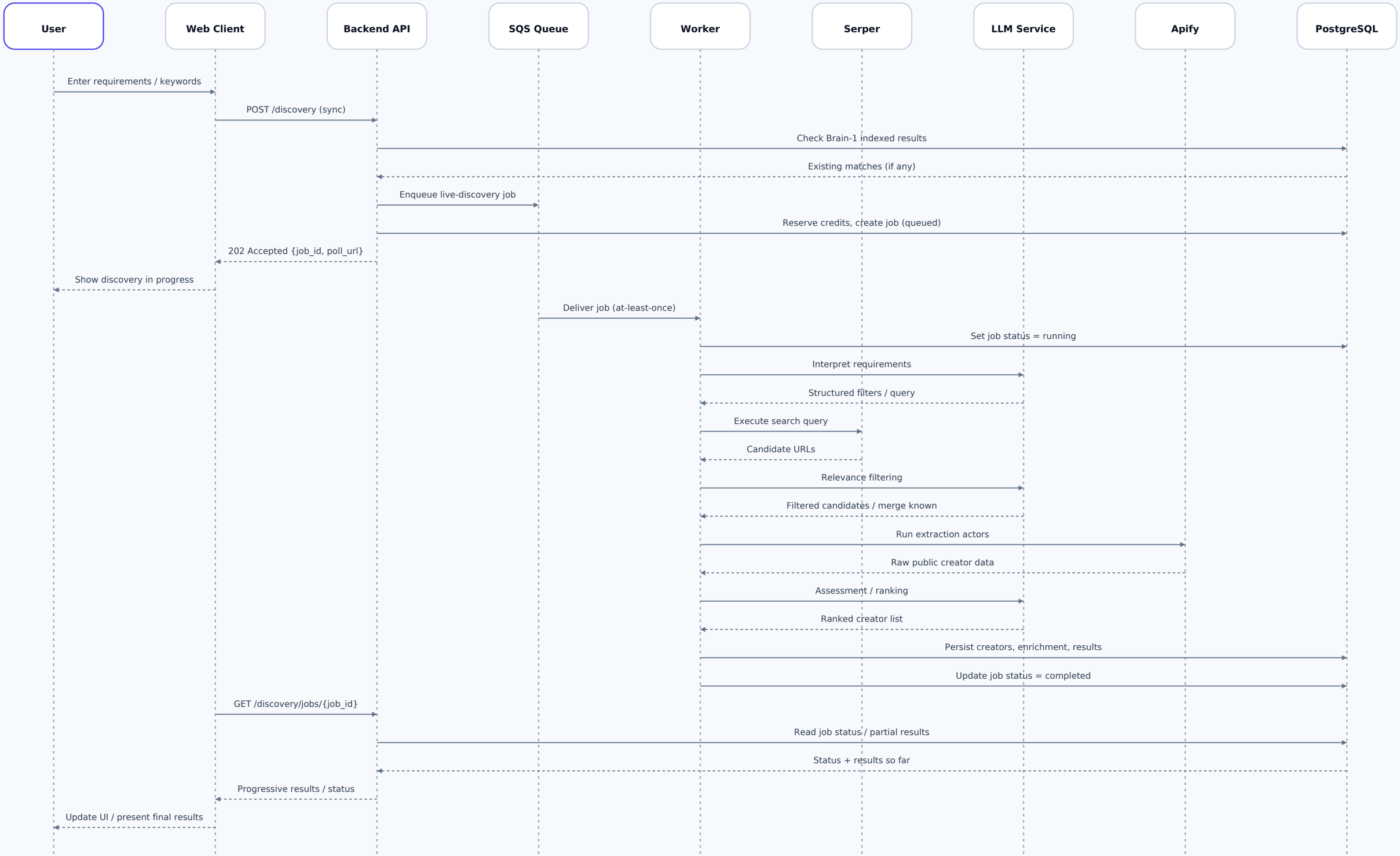


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7. Live Discovery Sequence Diagram

Asynchronous live discovery: synchronous job creation followed by queue/worker processing and client polling.

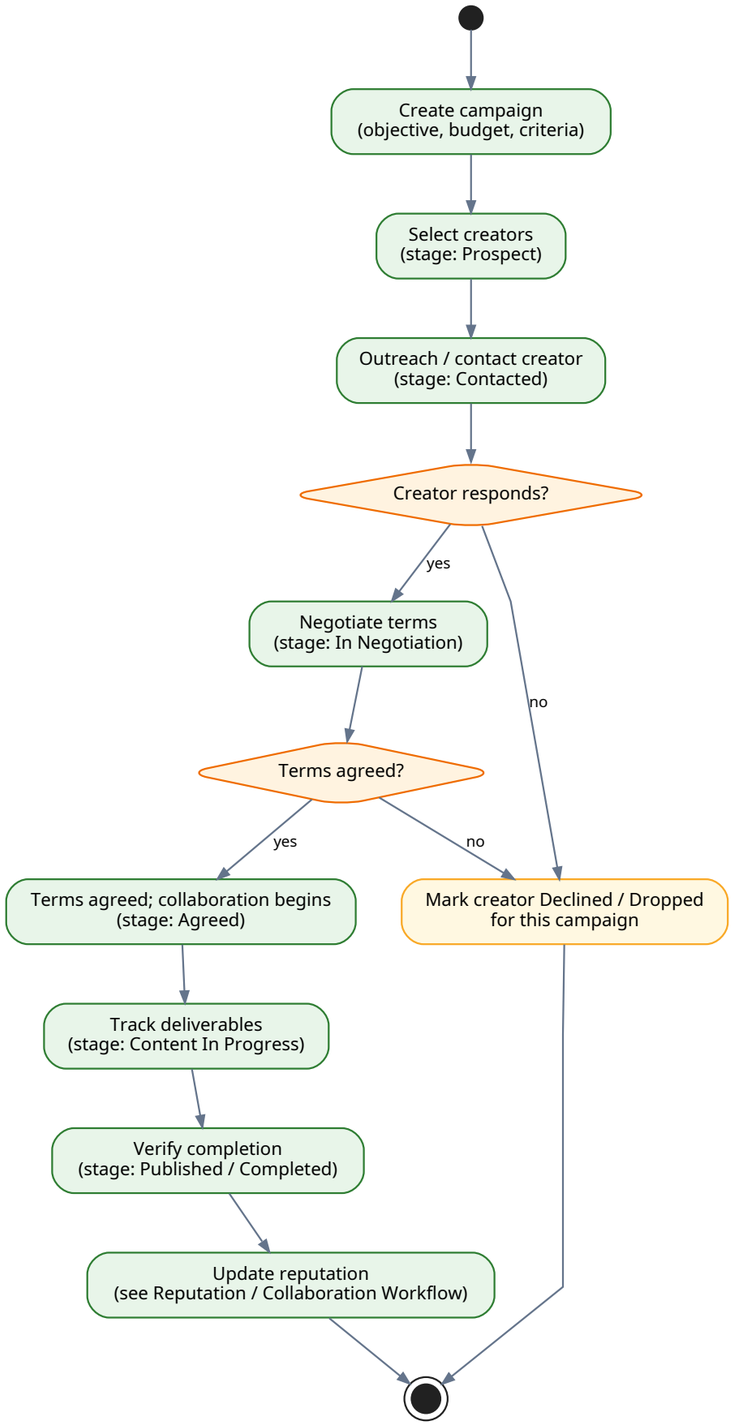


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8. Campaign Lifecycle Activity Diagram

Campaign workflow using the named MUSHIN pipeline stages; no payment/escrow step is introduced.

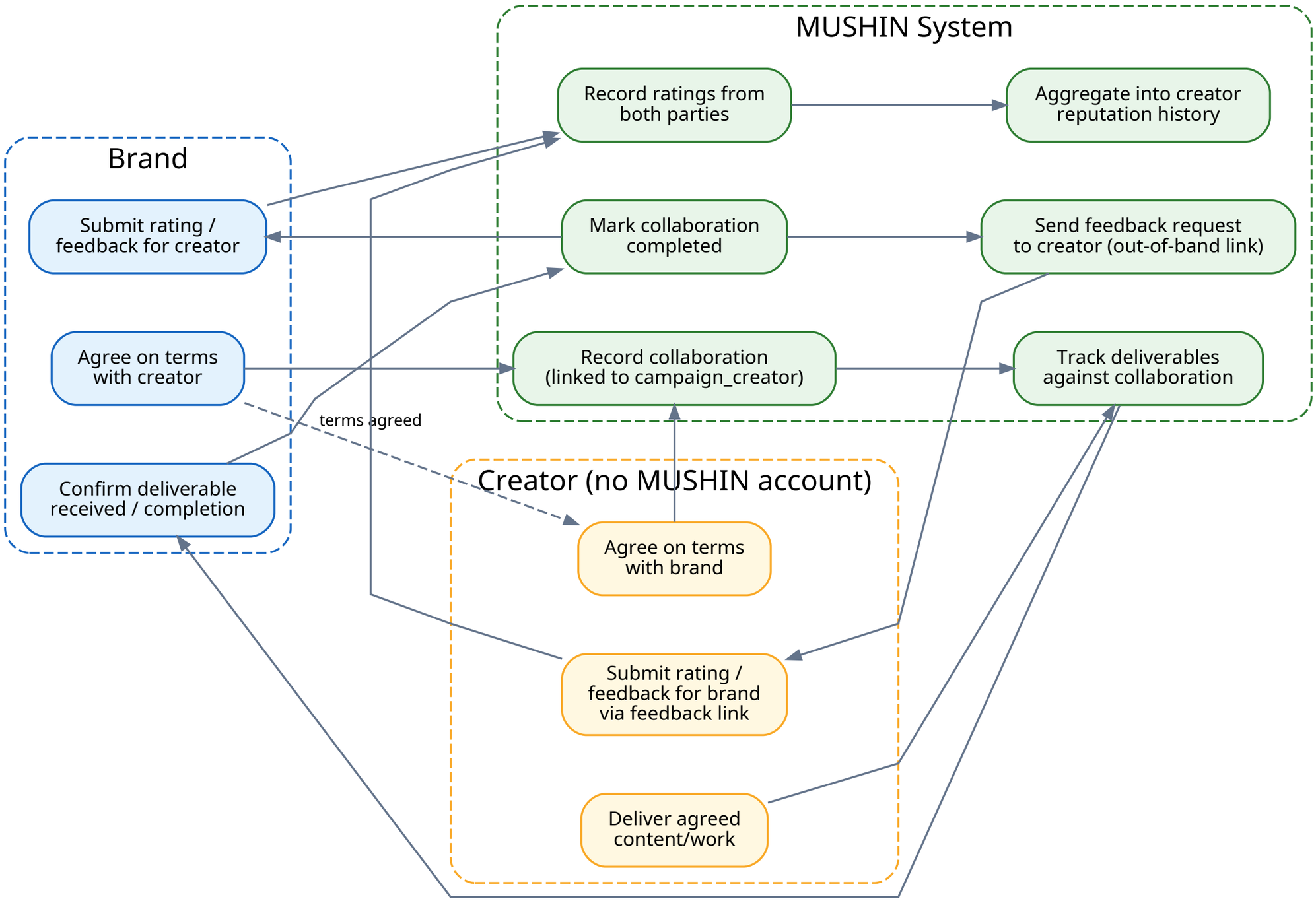


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9. Reputation / Collaboration Workflow

Bilateral collaboration and feedback flow represented as a distinct workflow.

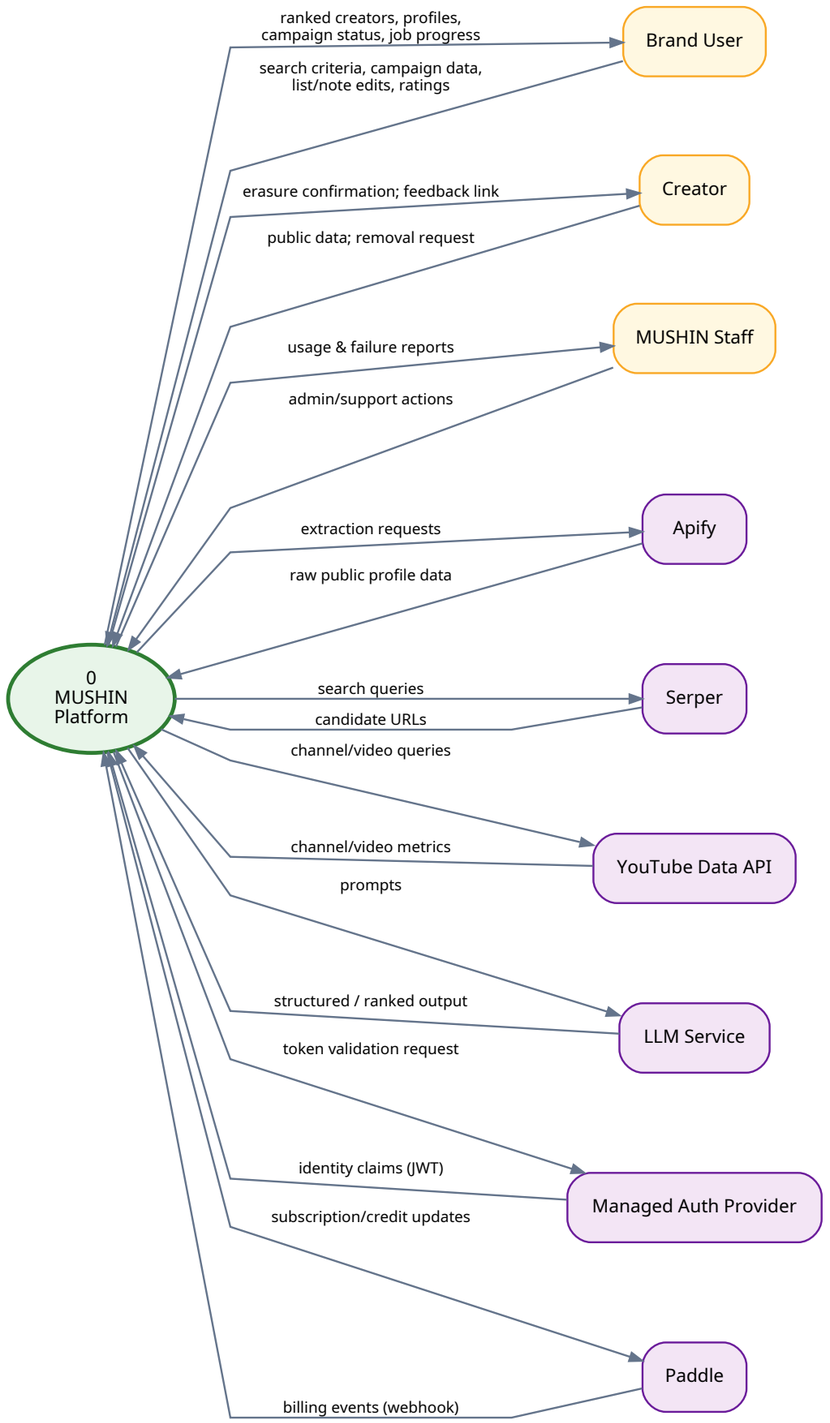


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10. DFD Level 0

MUSHIN as a single high-level process and its external data flows.

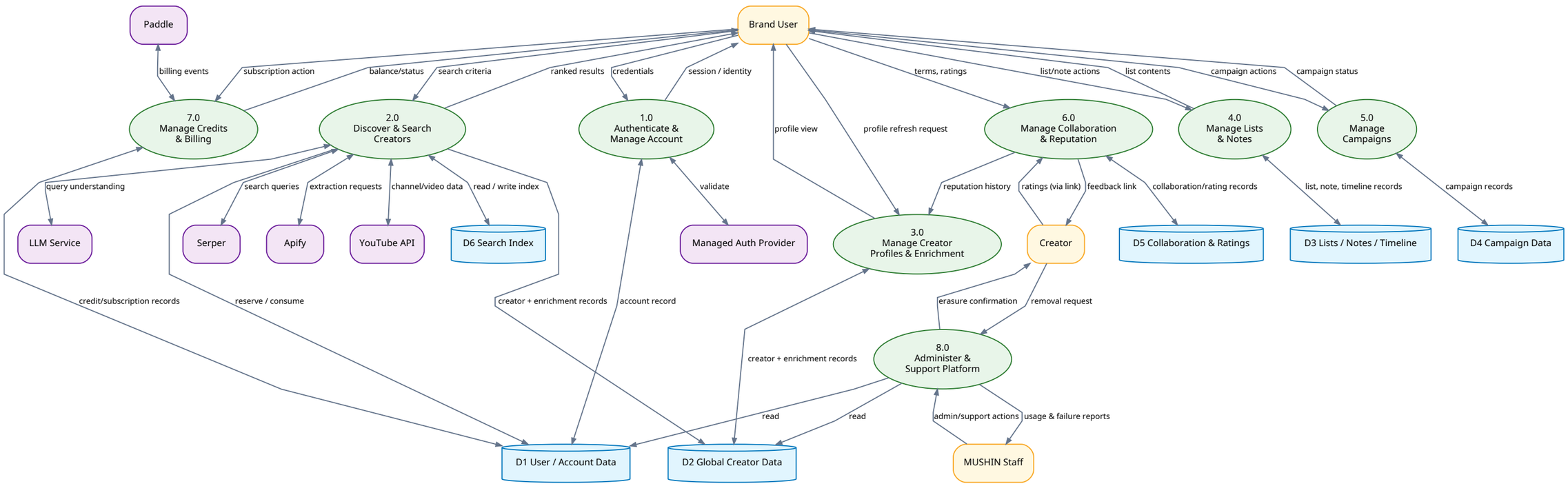


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11. DFD Level 1

Eight major MUSHIN processes and six logical data stores.

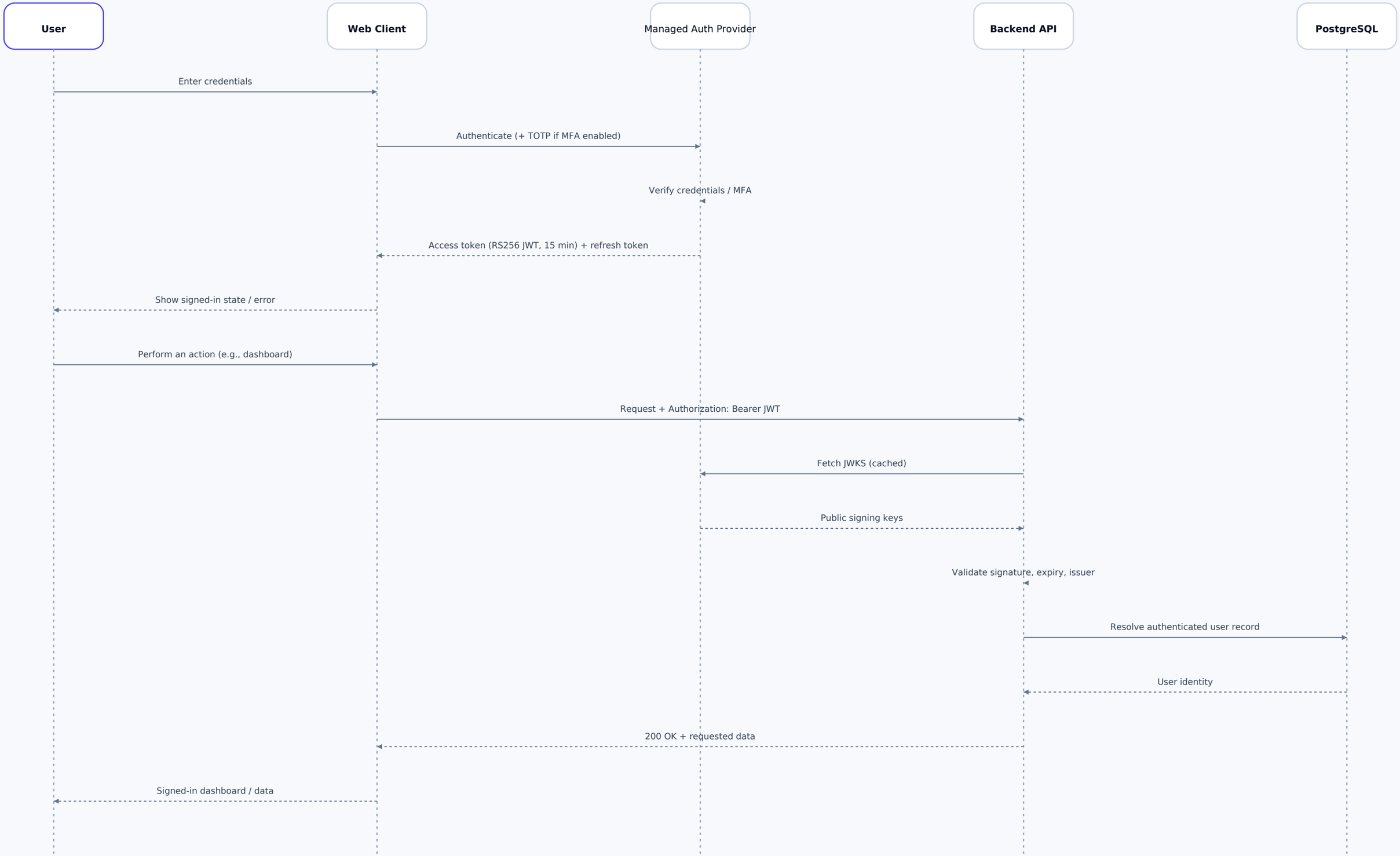


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Diagram rendered as vector artwork for readability

12. Authentication Sequence Diagram

Customer authentication flow using the managed identity provider, RS256 JWTs, JWKS validation, and user resolution.

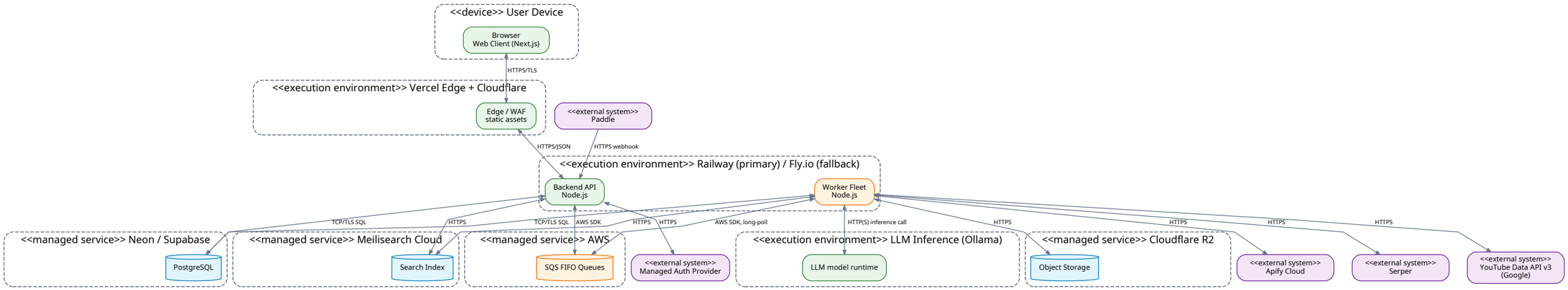


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UML sequence rendered for readability

13. Deployment Diagram

Runtime placement across user device, edge, application/worker infrastructure, data services, and external systems.



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Diagram rendered as vector artwork for readability

Diagram Consistency Notes

1. Workspace concept

The source document states that Workspace was removed for this diagram set. The diagrams therefore use a generalized User-owned model rather than reintroducing WORKSPACE.

2. LLM / Ollama vs hosted LLM tiers

The source retains its own stated distinction: vendor-neutral LLM / AI Service in some high-level diagrams, LLM Service (Ollama) in selected technical diagrams, and a dedicated Ollama execution environment in deployment. This PDF preserves that source terminology without altering it.

3. Reputation / collaboration scope

The source explicitly notes that the full reputation system is outside the Version 1.0 SRS scope, but the request included a concrete collaboration/rating workflow. COLLABORATION and RATING are therefore retained exactly as documented in the supplied Markdown.

4. Creator interaction

The source keeps Creator as a data subject without a MUSHIN account. The rating workflow uses an out-of-band feedback link rather than inventing a new authentication mechanism.

5. Outreach/contact management

Outreach is shown because it is included in the supplied use-case and campaign workflow material. The source notes that its interfaces are architecturally accommodated even though the SRS functional requirements did not fully define an outreach subsystem.

6. Staff authentication

Customer authentication is shown in detail; the source only references the separate staff identity realm rather than defining a second full sequence.

7. GCP/WP schema detail

The source says gcp/wp are implementation-level schema groupings and that cross-plane relationships are application-level. The ERD keeps the logical entity relationships readable rather than reproducing schema prefixes on every entity.

8. Conversion method

The uploaded Markdown contains Mermaid definitions. Because a dedicated Mermaid renderer was not available in the local conversion environment, the PDF uses equivalent vector diagram artwork generated from those same diagram definitions. No system functionality was intentionally changed for the conversion.